

A Degenerate Full Resolution of the Stated Non-Archimedean Zauner Problem

Candidate solution packet for arXiv:2210.07062

9 August 2026

Status: candidate full solution, likely valid. The result resolves Question 3.1 and Conjecture 3.2 exactly as written. Its mechanism is degenerate: the source does not require the vectors, or the lines they span, to be distinct. A distinct-line strengthening is not addressed.

1 Source and exact targets

The source is K. Mahesh Krishna, *Non-Archimedean Welch Bounds and Non-Archimedean Zauner Conjecture*, arXiv:2210.07062 (2022) [1]. Equation (2) of the source assumes that the complete non-Archimedean valued field $\mathbb{K}$ satisfies

$$\left| \sum_{j=1}^r \lambda_j^2 \right| = \max_{1 \leq j \leq r} |\lambda_j|^2 \quad (\lambda_1, \dots, \lambda_r \in \mathbb{K}, r \in \mathbb{N}). \quad (1)$$

The standard bilinear inner product on $\mathbb{K}^d$ is $\langle x, y \rangle = \sum_{i=1}^d x_i y_i$.

Question 3.1 asks for which $(d, n) \in \mathbb{N}^2$ there are vectors $\tau_1, \dots, \tau_n \in \mathbb{K}^d$ such that

- (i) $\langle \tau_j, \tau_j \rangle = 1$ for every j ;
- (ii) $S_\tau x = \sum_{j=1}^n \langle x, \tau_j \rangle \tau_j$ is diagonalizable;
- (iii)

$$\max_{j \neq k} \{ |n|, |\langle \tau_j, \tau_k \rangle|^2 \} = \frac{|n|^2}{|d|}. \quad (2)$$

The literal maximum is nonempty for $n \geq 2$. For $n = 1$ we use the natural reading

$$\max \{ |n|, \max_{j \neq k} |\langle \tau_j, \tau_k \rangle|^2 \}.$$

Conjecture 3.2 specializes to $n = d^2$. Its displayed condition (iii) uses an undefined symbol n ; the surrounding sentence makes $n = d^2$ the natural reading. The proof below also works if the intended right side was $|d|$.

Source evidence: Question 3.1 and Conjecture 3.2

Remark 2.4. Theorems [2,1] and [2,2] now by replacing $\mathbb{K}$ by a d -dimensional non-Archimedean power space over $\mathbb{K}$.

3. NON-ARCHIMEDEAN ZAUNER CONJECTURE AND OPEN PROBLEMS

Theorem [2.1] straight away brings the following question.

Question 3.1. *Given non-Archimedean field $\mathbb{K}$ satisfying Equation (2), for which $(d, n) \in \mathbb{N} \times \mathbb{N}$, there exist vectors $\tau_1, \dots, \tau_n \in \mathbb{K}^d$ satisfying the following.*

- (i) $\langle \tau_j, \tau_j \rangle = 1$ for all $1 \leq j \leq n$.
- (ii) The operator $S_\tau : \mathbb{K}^d \ni x \mapsto \sum_{j=1}^n \langle x, \tau_j \rangle \tau_j \in \mathbb{K}^d$ is diagonalizable.
- (iii)

$$\max_{1 \leq j, k \leq n, j \neq k} \{ |n|, |\langle \tau_j, \tau_k \rangle|^2 \} = \frac{|n|^2}{|d|}.$$

A particular case of Question [3.1] is the following non-Archimedean version of Zauner conjecture (see [2, 5, 10, 12, 33, 36, 43, 48, 51, 55, 63, 70, 84] for Zauner conjecture in Hilbert spaces, [53] for Zauner conjecture in Hilbert C^* -modules and [52] for Zauner conjecture in Banach spaces).

Conjecture 3.2. *(Non-Archimedean Zauner Conjecture) Let $\mathbb{K}$ non-Archimedean field satisfying Equation (2). For each $d \in \mathbb{N}$, there exist vectors $\tau_1, \dots, \tau_{d^2} \in \mathbb{K}^d$ satisfying the following.*

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The conjecture continues at the page break:

NON-ARCHIMEDEAN WELCH BOUNDS AND NON-ARCHIMEDEAN ZAUNER CONJECTURE

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- (i) $\langle \tau_j, \tau_j \rangle = 1$ for all $1 \leq j \leq d^2$.
 - (ii) The operator $S_\tau : \mathbb{K}^d \ni x \mapsto \sum_{j=1}^{d^2} \langle x, \tau_j \rangle \tau_j \in \mathbb{K}^d$ is diagonalizable.
 - (iii)

$$|\langle \tau_j, \tau_k \rangle|^2 = |n|, \quad \forall 1 \leq j, k \leq d^2, j \neq k.$$

We remember the definition of Gerzon's bound which allows us to recall the bounds which are in the

2 Proof intuition

The field hypothesis already collapses the numerical part of every Welch equality: substituting r copies of 1 into (1) gives $|r| = 1$ for every positive integer r . The source asks only for a *collection* of vectors and imposes no distinctness or spanning condition. We may therefore repeat a single unit vector. Taking $\tau_j = e_1$ makes every cross inner product equal to 1 and makes the frame operator the already diagonal matrix $\text{diag}(n, 0, \dots, 0)$. Both sides of (2) are then 1.

3 Formal result and proof

Lemma 1 (Positive integers are units in absolute value). *If $\mathbb{K}$ satisfies (1), then $|r| = 1$ for every $r \in \mathbb{N}$.*

Proof. In (1), set $\lambda_1 = \dots = \lambda_r = 1$. Then

$$|r| = |1^2 + \dots + 1^2| = \max_{1 \leq j \leq r} |1|^2 = 1.$$

□

Theorem 1 (Complete answer to Question 3.1). *Let $\mathbb{K}$ satisfy (1). For every $d \in \mathbb{N}$ and every $n \geq 2$, the requirements of Question 3.1 are satisfied. The same holds for $n = 1$ under the natural empty-off-diagonal convention described above.*

Proof. Let $e_1 = (1, 0, \dots, 0) \in \mathbb{K}^d$ and set

$$\tau_1 = \tau_2 = \dots = \tau_n = e_1.$$

First, $\langle \tau_j, \tau_j \rangle = \langle e_1, e_1 \rangle = 1$. For $x = (x_1, \dots, x_d)$,

$$S_\tau x = \sum_{j=1}^n \langle x, e_1 \rangle e_1 = nx_1 e_1.$$

Thus the matrix of S_τ in the standard basis is $\text{diag}(n, 0, \dots, 0)$, so it is diagonalizable (indeed, already diagonal). Finally, for every $j \neq k$,

$$|\langle \tau_j, \tau_k \rangle|^2 = |\langle e_1, e_1 \rangle|^2 = 1.$$

Lemma 1 gives $|n| = |d| = 1$, hence

$$\max_{j \neq k} \{|n|, |\langle \tau_j, \tau_k \rangle|^2\} = \max\{1, 1\} = 1 = \frac{1^2}{1} = \frac{|n|^2}{|d|}.$$

This is precisely the requested equality. $\square$

Corollary 1 (The stated non-Archimedean Zauner conjecture). *Conjecture 3.2 holds as written: for every $d \in \mathbb{N}$, take d^2 copies of e_1 . For $d \geq 2$ all three displayed conditions hold. For $d = 1$, the off-diagonal universal condition is vacuous. If the undefined $|n|$ in condition (iii) is read as either $|d^2|$ or the likely intended $|d|$, its value is 1 by Lemma 1.*

Corollary 2 (Simultaneous equality in every stated Welch order). *For the same repeated collection and every tensor order $m \in \mathbb{N}$, the higher-order bound of Theorem 2.3 in the source is an equality.*

Proof. The induced operator on $\text{Sym}^m(\mathbb{K}^d)$ is n times the coordinate projection onto $e_1^{\otimes m}$, hence diagonalizable. Every cross term has absolute value 1. Lemma 1, applied also to the positive integer $\binom{d+m-1}{m}$, makes both sides of the higher-order bound equal to 1. $\square$

4 The stated equiangular-line problem has no finite Gerzon bound

Question 3.6 asks for a maximum number of vectors satisfying only fixed self-inner-product and fixed cross-inner-product conditions. Its statement, which spans source pages 5–6, is reproduced below.

equiangular line problem using that).

Question 3.6. (Non-Archimedean Equiangular Line Problem) *Let $\mathbb{K}$ be a non-Archimedean field. Given $a \in \mathbb{K}$, $d \in \mathbb{N}$ and $\gamma > 0$, what is the maximum $n = n(\mathbb{K}, a, d, \gamma) \in \mathbb{N}$ such that there exist vectors $\tau_1, \dots, \tau_n \in \mathbb{K}^d$ satisfying the following.*

- (i) $\langle \tau_j, \tau_j \rangle = a$ for all $1 \leq j \leq n$.
- (ii) $|\langle \tau_j, \tau_k \rangle|^2 = \gamma$ for all $1 \leq j, k \leq n, j \neq k$.

In particular, whether there is a non-Archimedean Gerzon bound?

Question 3.6 can be easily modified to formulate question of non-Archimedean regular s -distance sets.

Corollary 3. *In the vector formulation of Question 3.6, $n(\mathbb{K}, 1, d, 1)$ is unbounded for every $d \in \mathbb{N}$. Consequently no finite non-Archimedean Gerzon bound can hold under the displayed hypotheses alone.*

Proof. For every prescribed $N \in \mathbb{N}$, take N copies of e_1 . Then every self-inner-product is 1 and every cross-inner-product has squared absolute value 1. Thus admissible collections exist with arbitrarily large cardinality. $\square$

5 Verification notes and limitations

The argument is purely algebraic and has no numerical or asymptotic dependency. The proof audit checked the following points.

- Equation (1) is quantified over every positive integer, so the integer-unit substitution is legitimate for n , d , and the binomial dimensions in the higher-order result.
- The source’s inner product is bilinear, and $\langle e_1, e_1 \rangle = 1$.
- The displayed frame operator is exactly diagonal in the standard basis.
- Neither Question 3.1, Conjecture 3.2, nor the vector conditions in Question 3.6 require distinct vectors, distinct one-dimensional subspaces, spanning, injectivity, or a tight frame operator.

This last point is the essential scope limitation. If “Zauner” or “equiangular lines” is strengthened to require d^2 distinct lines, or if S_τ is required to be a nonzero scalar multiple of the identity, the repeated-vector construction is inadmissible and this packet makes no claim. The result is full only for the exact formal statements in the source.

6 Bounded novelty check and review recommendation

On 9 August 2026 we searched the run’s registry, solution, attempt, and proof-gap indexes for arXiv:2210.07062 and the core phrases “Non-Archimedean Zauner Conjecture”, “Non-Archimedean Welch bounds”, and “non-Archimedean equiangular line problem”. We also performed bounded official-arXiv web searches by exact title, arXiv id, exact conjecture phrase, “distinct”, and “solution”. The source and adjacent classical SIC papers were found, but no later amendment adding distinctness and no matching resolution by repeated vectors was located. Novelty is therefore provisional and requires expert bibliographic review.

Human-review recommendation. Verify only two semantic points: that “collection” is to be read literally as allowing repetitions, and that the run evaluates open questions as stated rather than silently importing the classical distinct-line/tight-frame definition. Under that literal reading, the proof is complete.

Reference. [1] K. Mahesh Krishna, *Non-Archimedean Welch Bounds and Non-Archimedean Zauner Conjecture*, arXiv:2210.07062, 2022, <https://arxiv.org/abs/2210.07062>.